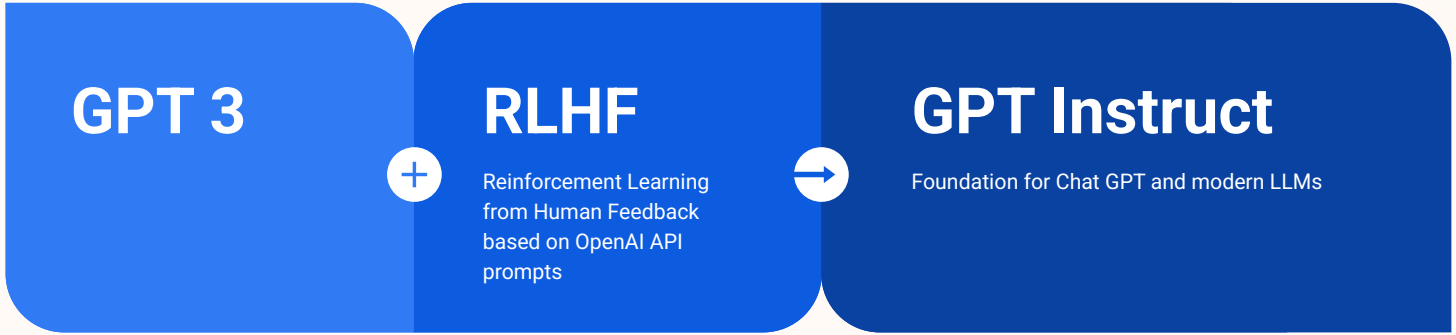
Aligning LLMs:
Instruction Following

01

Training language models to follow instructions with human feedback

Authors: Ouyang et al. (OpenAI)
Date: 2022

Abstract


GPT 3
RLHF
Reinforcement Learning from Human Feedback based on OpenAI API prompts
GPT Instruct
Foundation for Chat GPT and modern LLMs

The main problem: Completion vs Answer

Prompt:
What is the capital of France?

The main problem: Completion vs Answer

Prompt:
What is the capital of France?

Response:

A. Berlin
B. London
C. Paris
D. Wrocław

The main problem: Completion vs Answer

Prompt:
What is the capital of France?

Response:

A. Berlin
B. London
C. Paris
D. Wrocław

Prompt:
Explain the moon landing to a 6 year old.

Reponse:
"Explain the theory of gravity to a 6 year old."
"Explain the theory of relativity to a 6 year old."
"Explain the Big Bang to a 6 year old."

The Goal

Model should be:

- Helpful
- Honest
- Harmless

Fine-tuning Methods

Dataset

- Prompts written by labelers and prompts from the OpenAI API playground (mostly in English).
- diverse tasks including generation, question answering, dialog, summarization, extractions, coding, and more.
- The dataset format: (Prompt, Response) pairs

Labelers

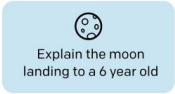
Contractors from some external company that passed some screening tests.

Fine-tuning Methods

Step 1

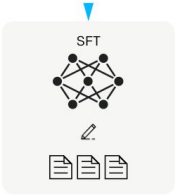
Collect demonstration data, and train a supervised policy.

A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.

This data is used to fine-tune GPT-3 with supervised learning.



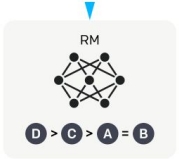
Step 2

Collect comparison data, and train a reward model.

A prompt and several model outputs are sampled.

A labeler ranks the outputs from best to worst.

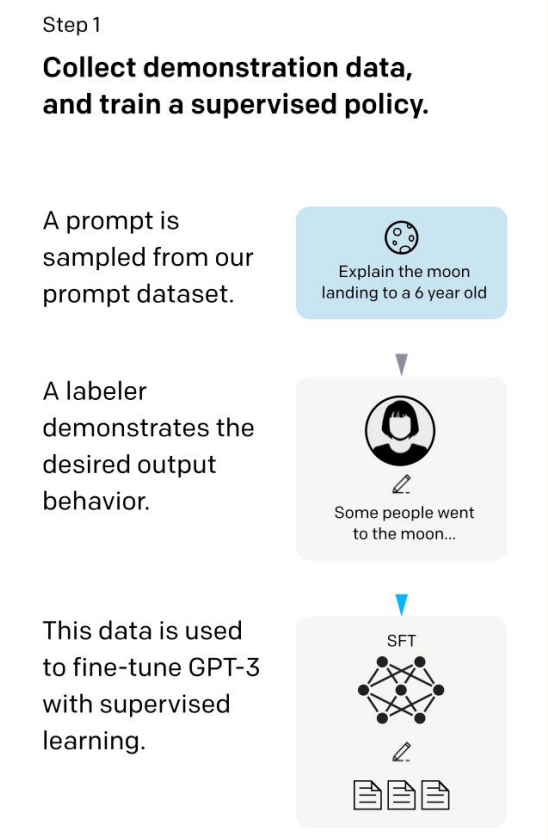
This data is used to train our reward model.



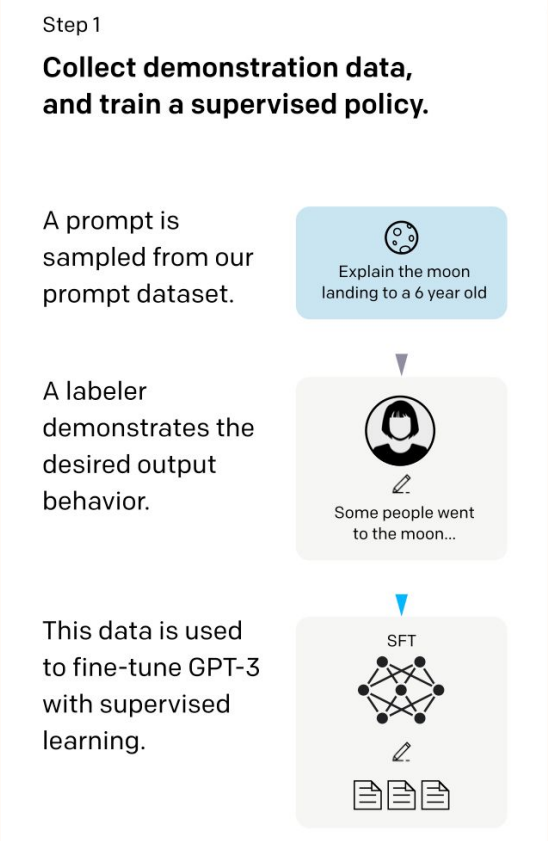
Step 3

Optimize a policy against the reward model using reinforcement learning.

Fine-tuning Methods - Step 1: Baseline



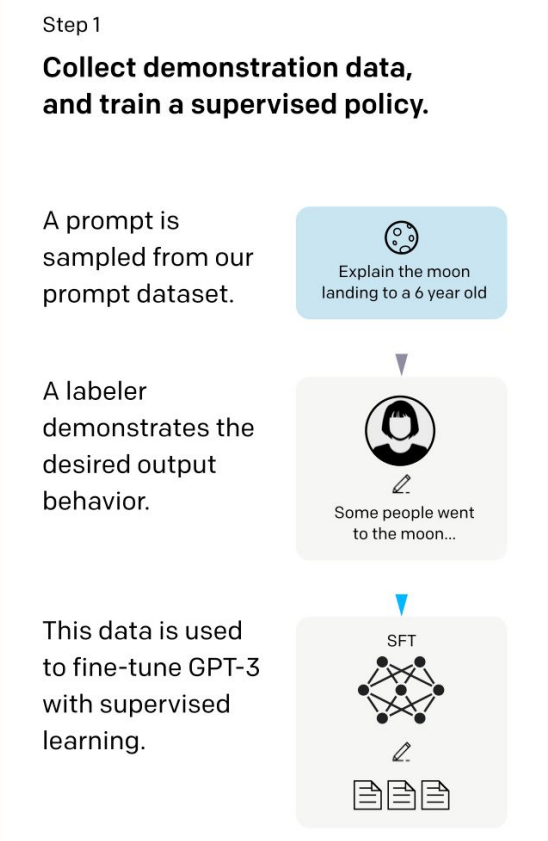
Fine-tuning Methods - Step 1: Baseline



SFT model overfits on purpose.

- Better results in following steps
- Better human preference scores

Fine-tuning Methods - Step 1: Baseline



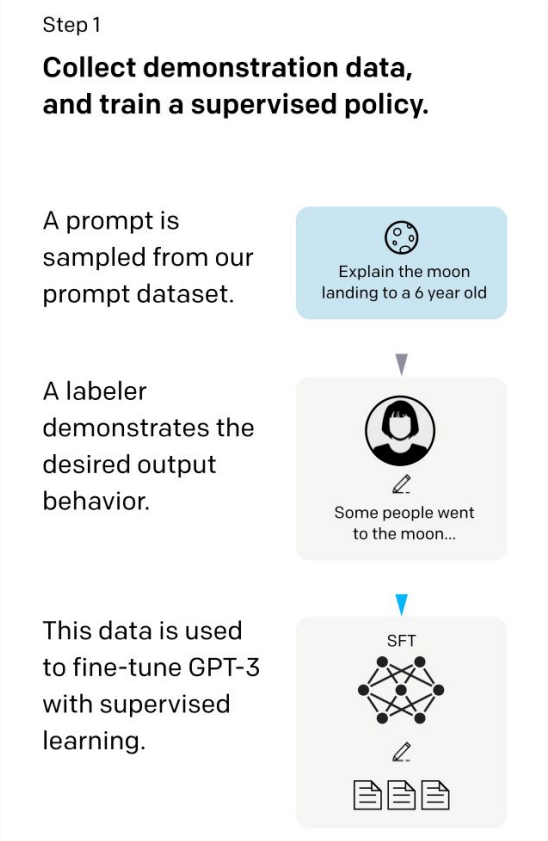
SFT model overfits on purpose.

- Better results in following steps
- Better human preference scores

Why not stop here?

- Model mimics the style of labelers
- People are slow

Fine-tuning Methods - Step 1: Baseline



SFT model overfits on purpose.

- Better results in following steps
- Better human preference scores

Why not stop here?

- Model mimics the style of labelers
- People are slow

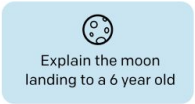
Step 1 output:
SFT (baseline model) – somewhat capable of following instructions

Fine-tuning Methods - Step 2: Scaling

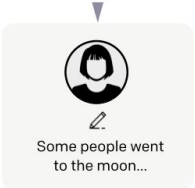
Step 1

Collect demonstration data, and train a supervised policy.

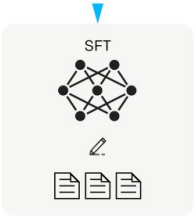
A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



This data is used to fine-tune GPT-3 with supervised learning.



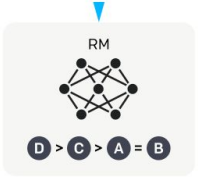
Step 2

Collect comparison data, and train a reward model.

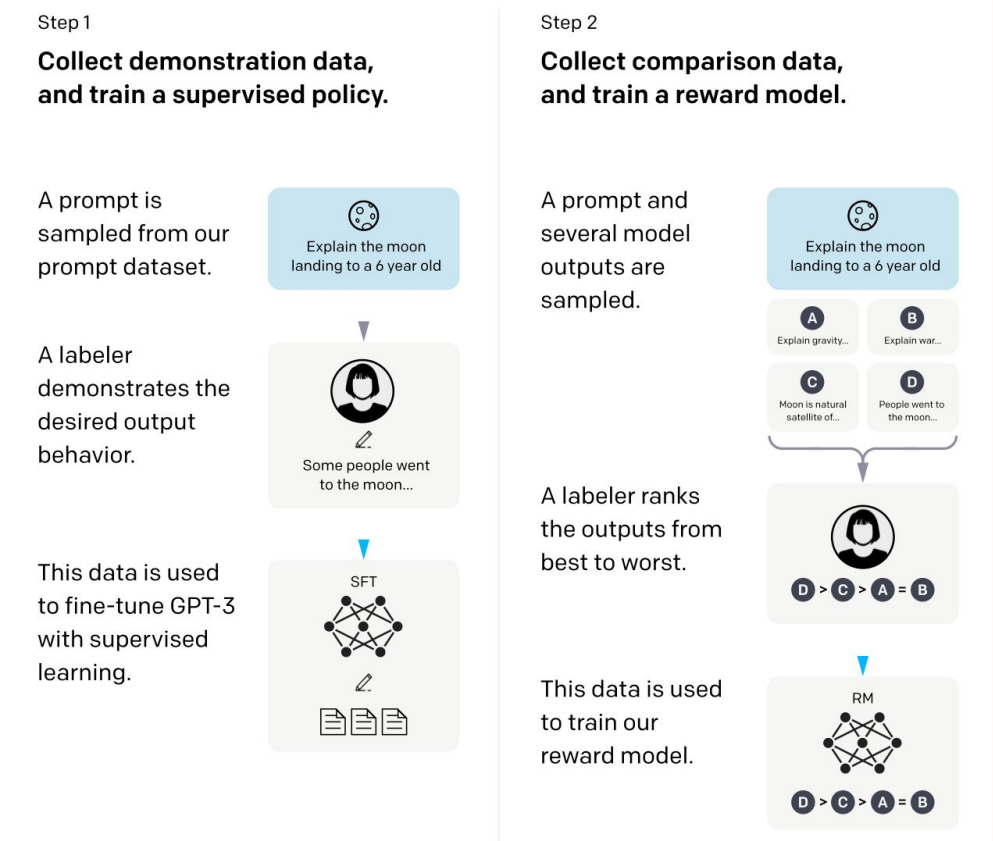
A prompt and several model outputs are sampled.

A labeler ranks the outputs from best to worst.

This data is used to train our reward model.

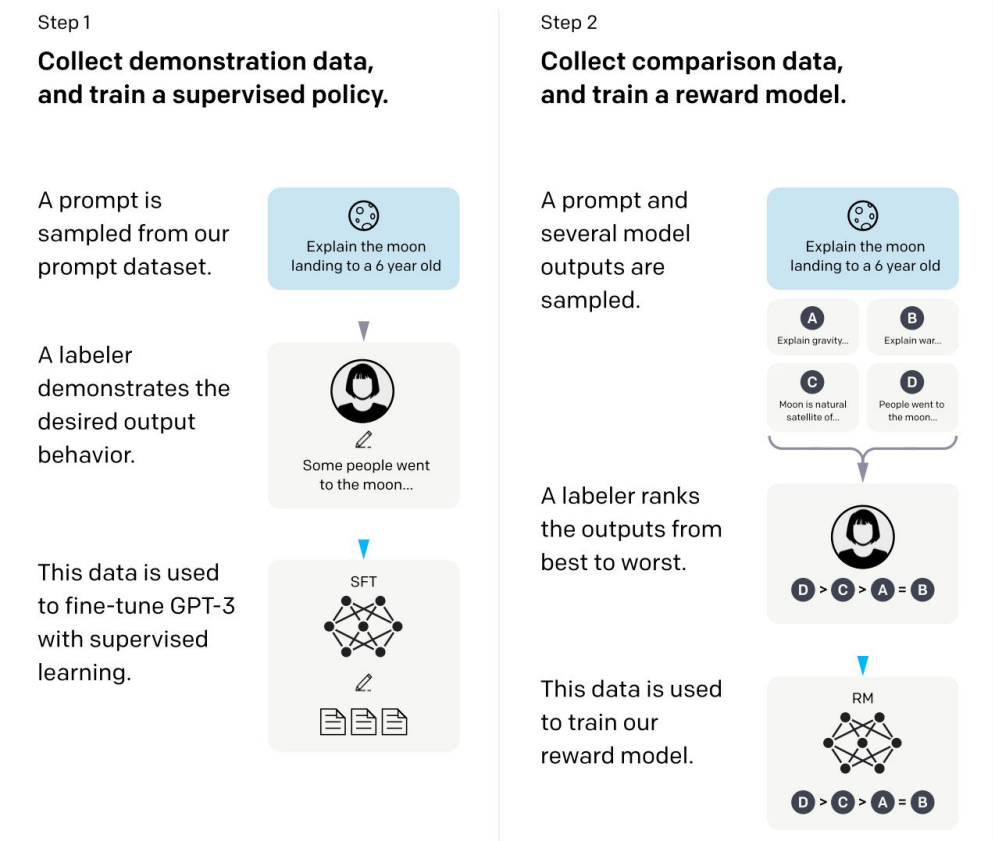


Fine-tuning Methods - Step 2: Scaling



Human feedback optimization:
Order instead of ranking

Fine-tuning Methods - Step 2: Scaling



Human feedback optimization:

Order instead of ranking

Step 2 output:

Reward model

Input: prompt & response

Output: scalar ranking of the response

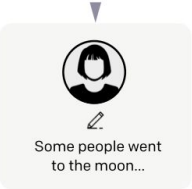
Fine-tuning Methods

Step 1

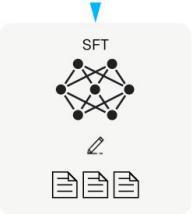
Collect demonstration data, and train a supervised policy.

A prompt is sampled from our prompt dataset.

A labeler demonstrates the desired output behavior.



This data is used to fine-tune GPT-3 with supervised learning.



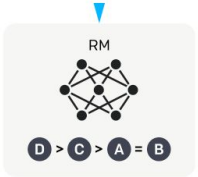
Step 2

Collect comparison data, and train a reward model.

A prompt and several model outputs are sampled.

A labeler ranks the outputs from best to worst.

This data is used to train our reward model.



Step 3

Optimize a policy against the reward model using reinforcement learning.

Fine-tuning Methods - Step 3 details

Step 3

Optimize a policy against the reward model using reinforcement learning.

RL Goal:
Maximize the objective function

$$\text{objective}(\phi) = E_{(x,y)\sim D_{\pi_\phi^{\mathrm{RL}}}}\left[r_\theta(x,y) - \beta \log\left(\pi_\phi^{\mathrm{RL}}(y \mid x)/\pi^{\mathrm{SFT}}(y \mid x)\right)\right] + \gamma E_{x\sim D_{\text{pretrain}}}\left[\log(\pi_\phi^{\mathrm{RL}}(x))\right]$$

Fine-tuning Methods - Step 3 details

Step 3

Optimize a policy against the reward model using reinforcement learning.

Reward – “Be helpful” term

$$\text{objective}(\phi) = E_{(x,y)\sim D_{\pi_\phi^{\mathrm{RL}}}}\left[\underline{r_\theta(x,y)} - \beta \log\left(\pi_\phi^{\mathrm{RL}}(y \mid x)/\pi^{\mathrm{SFT}}(y \mid x)\right)\right] + \gamma E_{x\sim D_{\text{pretrain}}}\left[\log(\pi_\phi^{\mathrm{RL}}(x))\right]$$

Fine-tuning Methods - Step 3 details

Step 3

Optimize a policy against the reward model using reinforcement learning.

Reward – "Be helpful" term

KL Penalty – "Stay normal" term

$$\text{objective}(\phi) = E_{(x,y)\sim D_{\pi_\phi^{\mathrm{RL}}}}\left[r_\theta(x,y) - \beta \log\left(\pi_\phi^{\mathrm{RL}}(y \mid x)/\pi^{\mathrm{SFT}}(y \mid x)\right)\right] + \gamma E_{x\sim D_{\text{pretrain}}}\left[\log(\pi_\phi^{\mathrm{RL}}(x))\right]$$

Fine-tuning Methods - Step 3 details

Step 3

Optimize a policy against the reward model using reinforcement learning.

Reward – "Be helpful" term

KL Penalty – "Stay normal" term

$$\text{objective}(\phi) = E_{(x,y)\sim D_{\pi_\phi^{\mathrm{RL}}}}\left[r_\theta(x,y) - \beta \log\left(\pi_\phi^{\mathrm{RL}}(y \mid x)/\pi^{\mathrm{SFT}}(y \mid x)\right)\right] + \gamma E_{x\sim D_{\text{pretrain}}}\left[\log(\pi_\phi^{\mathrm{RL}}(x))\right]$$

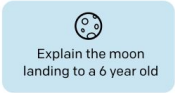
Pre Training – "Stay smart" term

Fine-tuning Methods - Pipeline summary

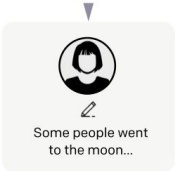
Step 1

Collect demonstration data, and train a supervised policy.

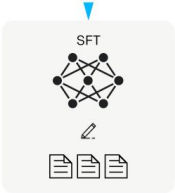
A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



This data is used to fine-tune GPT-3 with supervised learning.

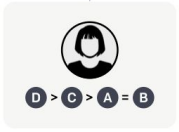


Step 2

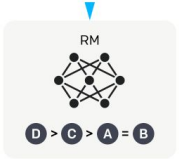
Collect comparison data, and train a reward model.

A prompt and several model outputs are sampled.

A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



Step 3

Optimize a policy against the reward model using reinforcement learning.

Evaluation

Model should be:

- Helpful

  Goal: Follow instructions & Infer intention from prompt

  Metric: Labeler preference

- Honest

  Goal: Model's statements should be true

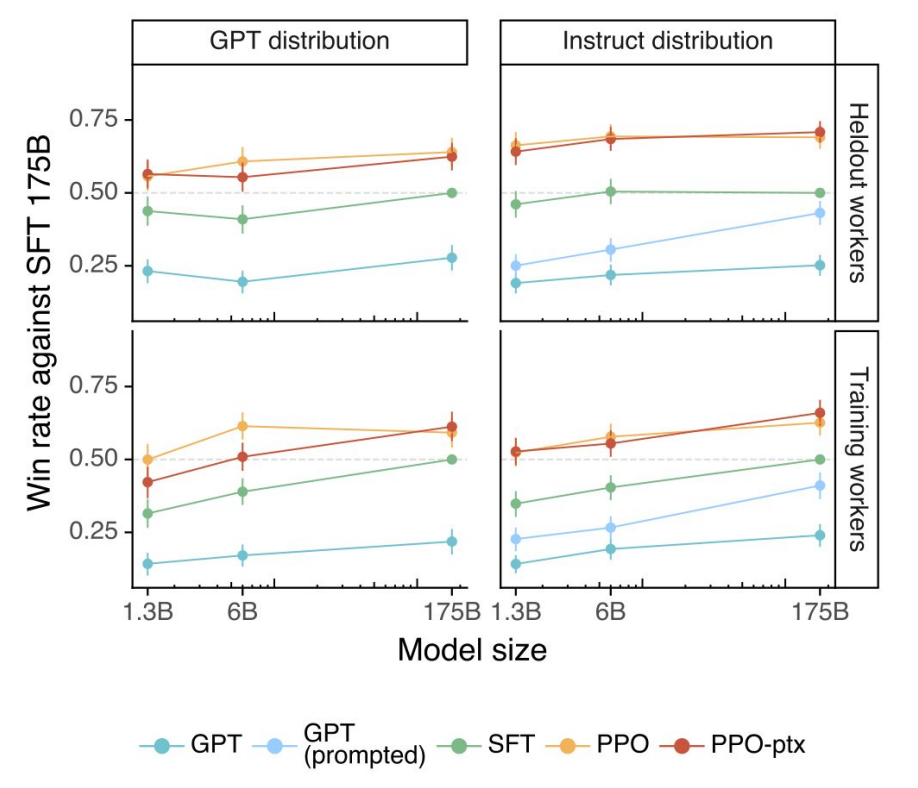
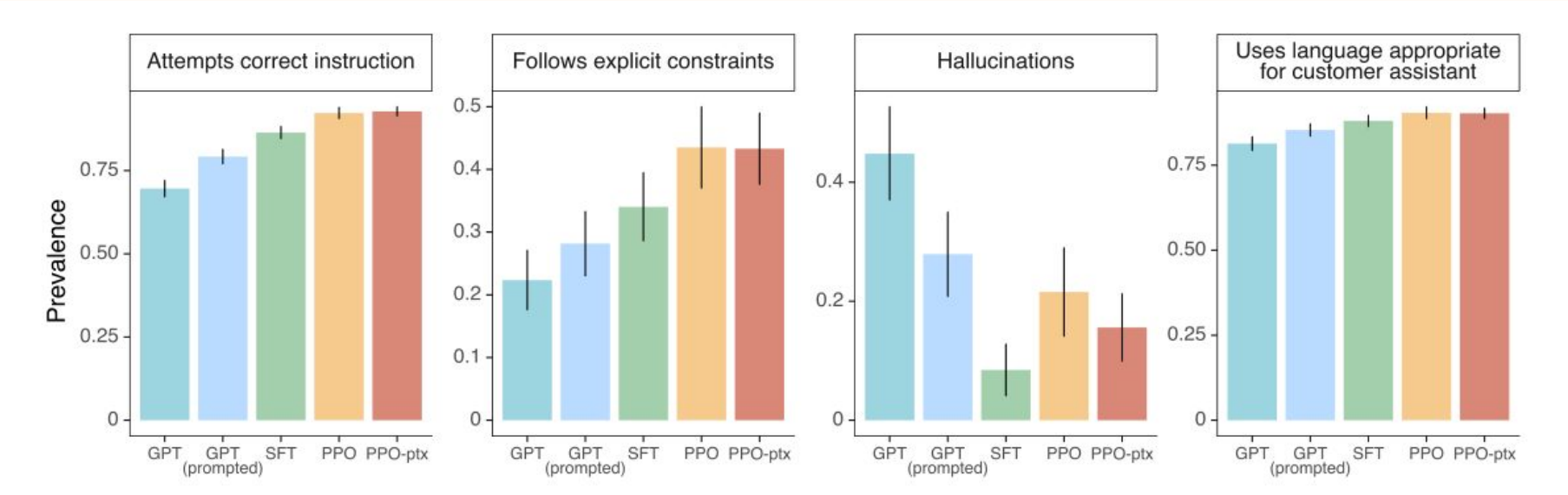
  Metric: evaluating tendency to hallucinate and answers on TruthfulQA dataset

- Harmless

  Goal: no inappropriate / racist / biased / sexual / violent output

  Metric: Labeler evaluation

Results - preferred response



Results - crucial requirements